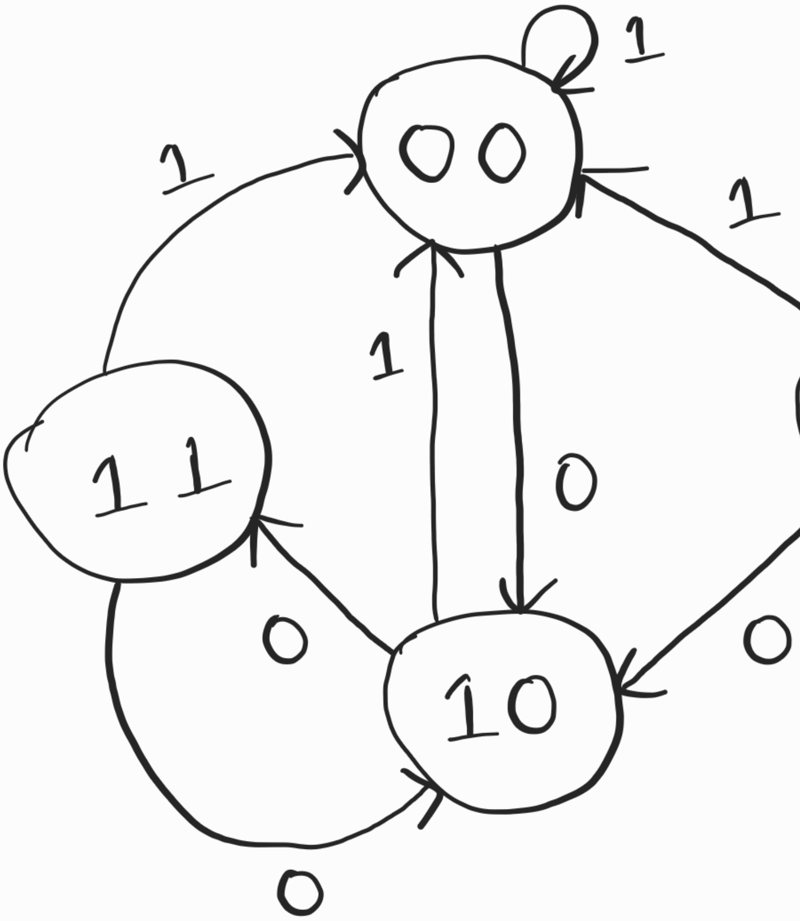
Q 1

$$J_A = A'x' \qquad K_A = A' + x$$

$$D_B = (A' + x + B)'$$
$$= AB'x'$$

A	B	x	J_A	K_A	D_B	A^+	B^+
0	0	0	1	1	0	1	0
0	0	1	0	1	0	0	0
0	1	0	1	1	0	1	0
0	1	1	0	1	0	0	0
1	0	0	0	0	1	1	1
1	0	1	0	1	0	0	0
1	1	0	0	0	0	1	0
1	1	1	0	1	0	0	0



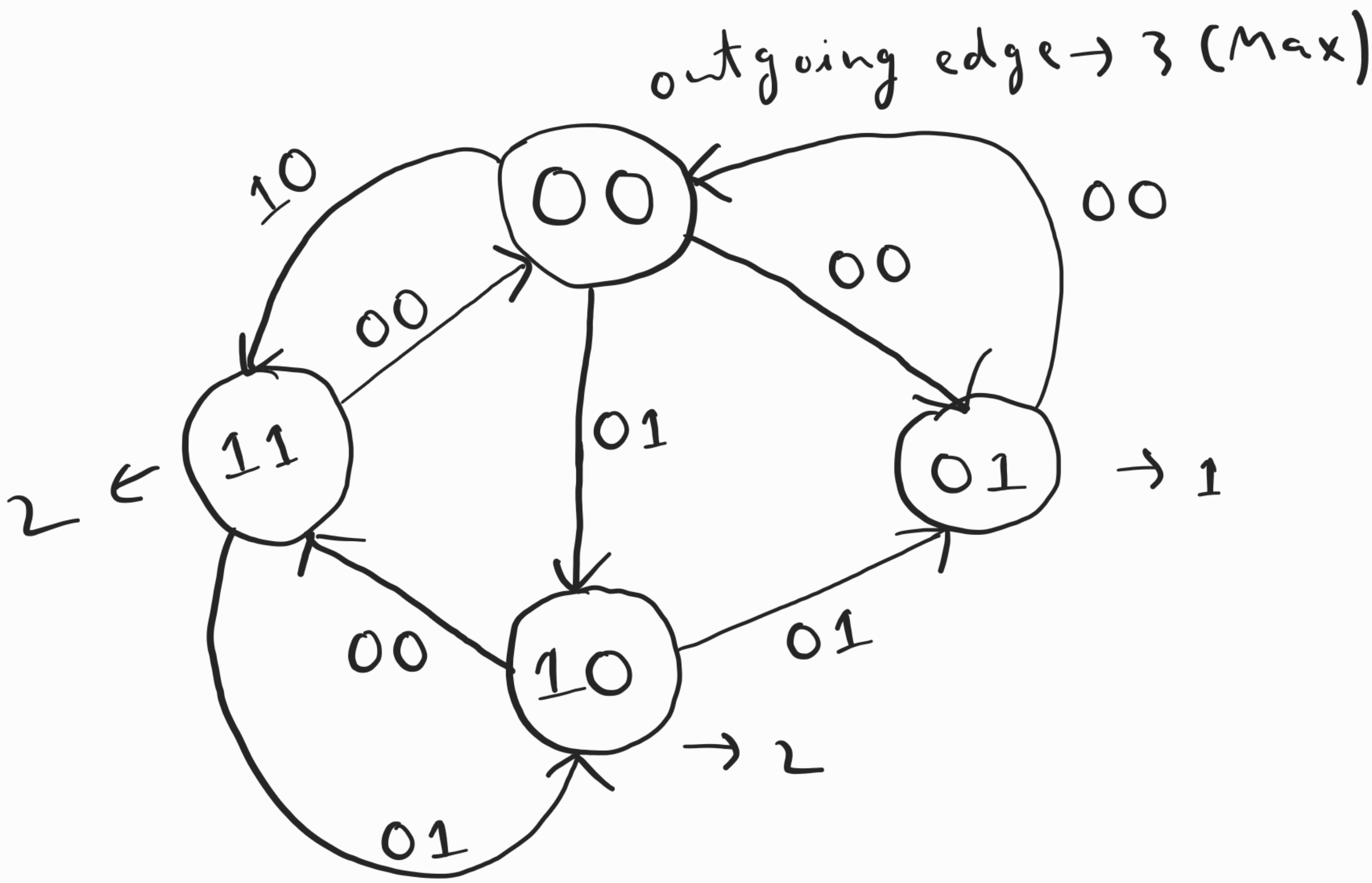
Q2

Leon → 00, Claire → 01, Ada → 10,
Chris → 11

No of states → 4

No of FF required → $\lceil \log_2 4 \rceil$
$= 2$

No of external inputs → $\lceil \log_2 3 \rceil$
$= 2$
(x and y)



Present states → A and B

Next states → A+ and B+

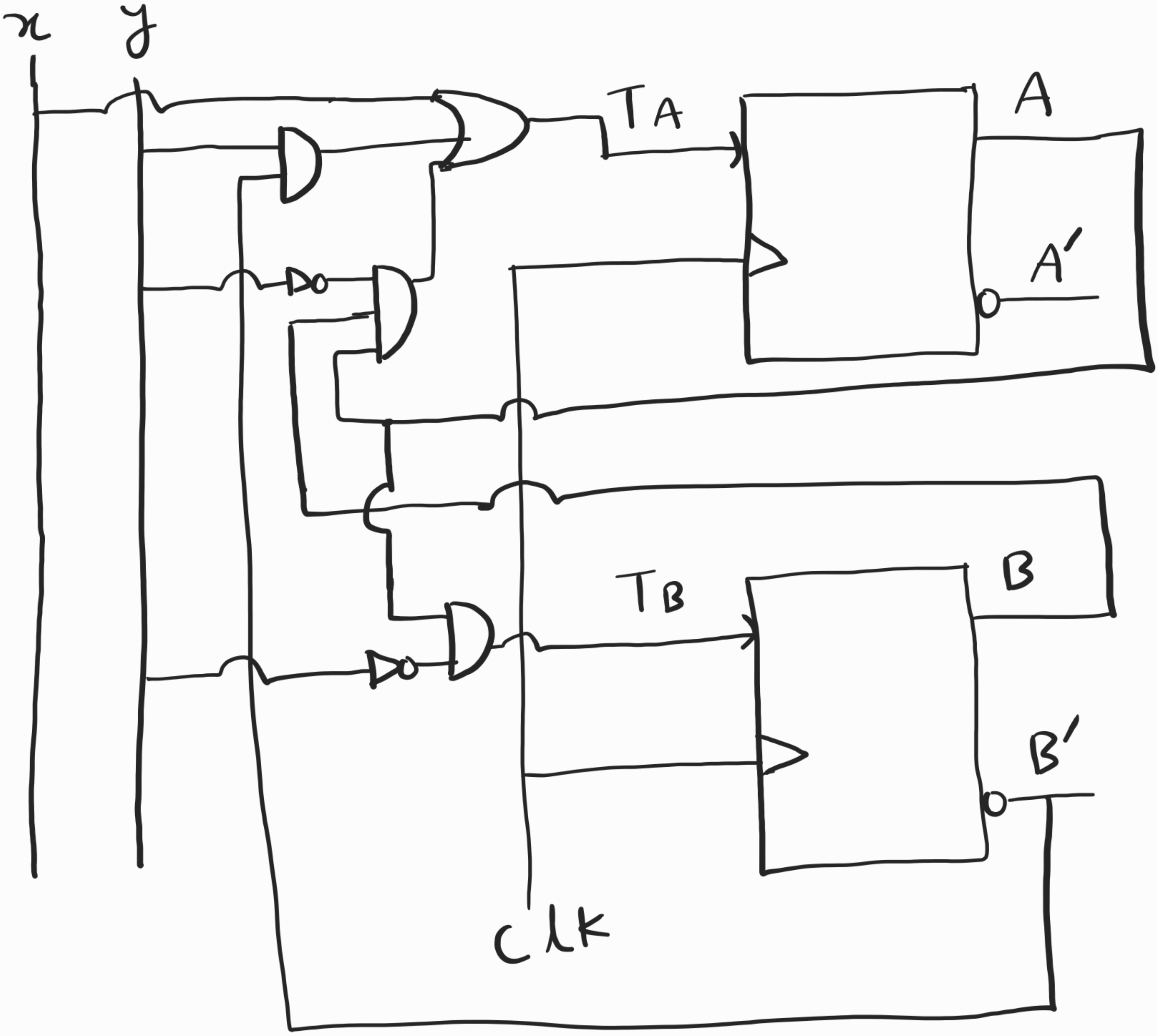
A	B	x	y	A^+	B^+	T_A	T_B
0	0	0	0	0	1	0	1
0	0	0	1	1	0	1	0
0	0	1	0	1	1	1	1
0	0	1	1	X	X	X	X
0	1	0	0	0	0	0	1
0	1	0	1	X	X	X	X
0	1	1	0	X	X	X	X
0	1	1	1	X	X	X	X
1	0	0	0	1	1	0	1
1	0	0	1	0	1	1	1
1	0	1	0	X	X	X	X
1	0	1	1	X	X	X	X
1	1	0	0	0	0	1	1
1	1	0	1	1	0	0	1
1	1	1	0	X	X	X	X
1	1	1	1	X	X	X	X

$$T_A = x + B'y + ABy'$$

$AB \backslash xy$	$x'y'$	$x'y$	xy	xy'
$A'B'$	(0)	(1) 1	(3) X	(2) 1
$A'B$	(4)	(5) X	(7) X	(6) X
AB	(12) 1	(13)	(15) X	(14) X
AB'	(8)	(9) 1	(11) X	(10) X

$$T_B = y' + A$$

$AB \backslash xy$	$x'y'$	$x'y$	xy	xy'
$A'B'$	(0) 1	(1)	(3) X	(2) 1
$A'B$	(4) 1	(5) X	(7) X	(6) X
AB	(12) 1	(13) 1	(15) X	(14) X
AB'	(8) 1	(9) 1	(11) X	(10) X

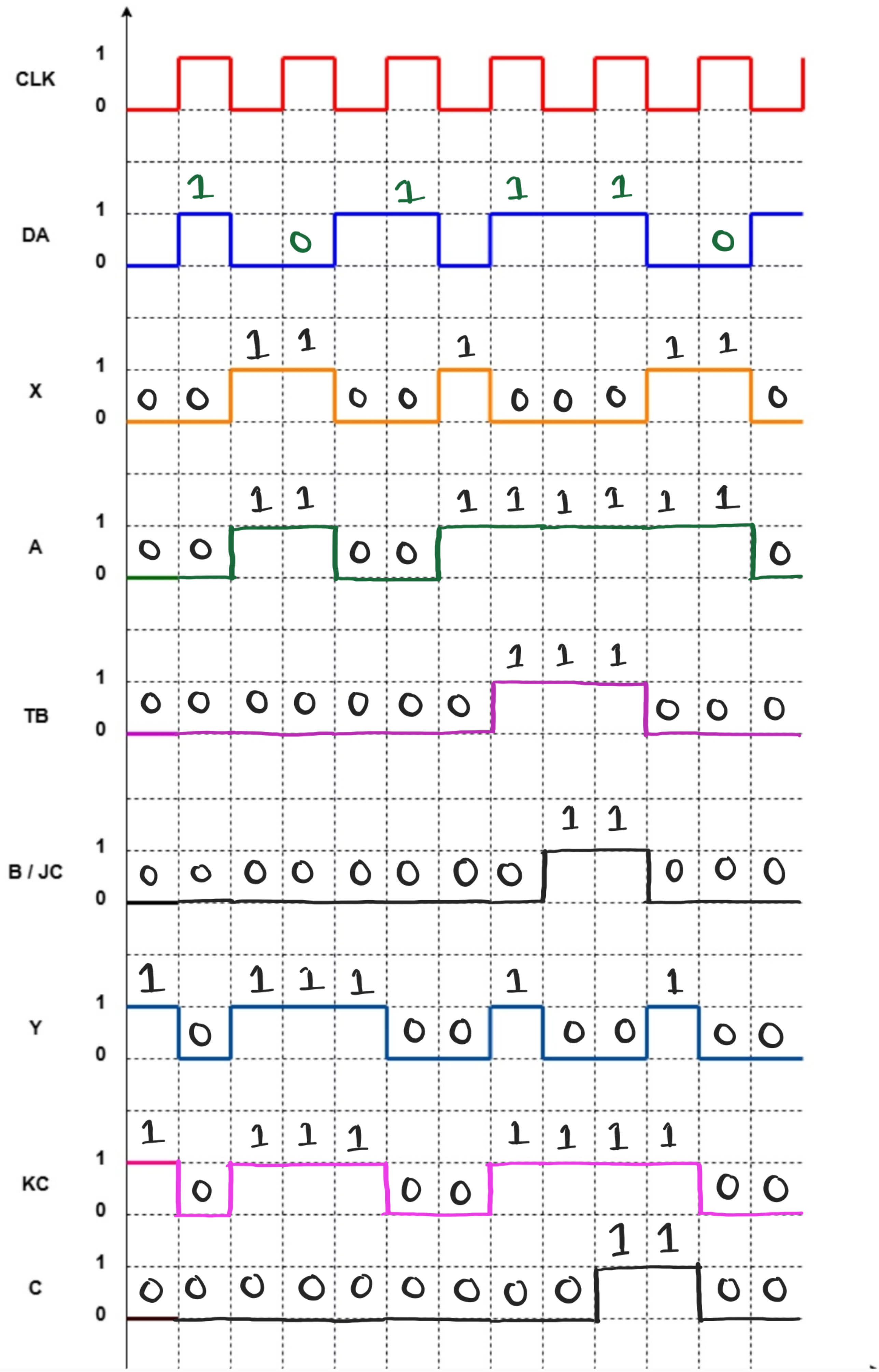


Q3

A → -ve edge

B → -ve edge

C → +ve edge


CLK
DA
X
A
TB
B / JC
Y
KC
C
1
0